Central Operations Case Study

Solution Report for 4 Problems

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1. Approach

I approached this case as a data analyst who is new to Revenue Cycle Management (RCM). RCM is the process that turns a patient visit into payment: the visit is copied from the clinic's system (Extraction), a claim is sent to the insurer (Submission), the insurer's decision is collected (Reconciliation), and failed claims are corrected (Denials).

I used the same method for every problem: look at the data or a sample of it, identify the pattern, group the reasons, fix the largest group first and then measure the result. I learned the domain rules needed for each problem (CPT codes, Medicare pricing, billing rules) as I went, verified every number with code, and state assumptions where the case does not provide data.

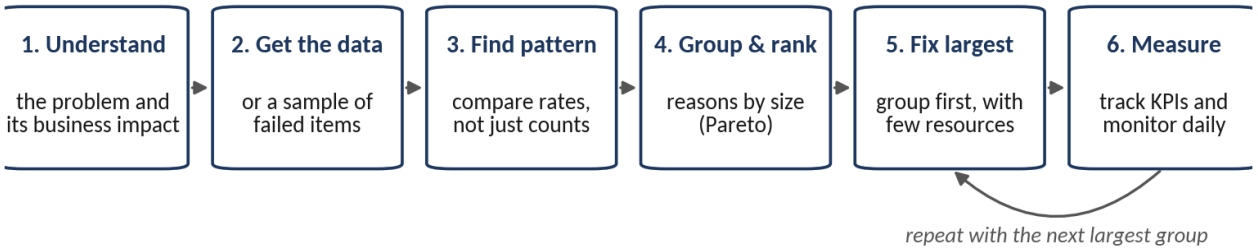


Figure 1. Approach applied to every problem.

2. Executive summary

Problem	Finding	Impact	Recommendation
P2 – Missing encounters	The import filter blocks visits containing non-CPT codes (e.g. NORCM, SP, TOS). 98.6% of these visits are missing versus 1.2% of other visits.	282 of 6,612 visits (4.26%) not imported; 202 blocked visits also contain billable codes.	Confirm the codes with the client, route such visits to a review queue, and run a daily EHR-to-Normandy check.
P4 – Debugging revenue	Payment differences are explained by location indexes (A), the multiple-procedure reduction (B) and incorrect units (C).	Claim C billed 3 units of an untimed code and missed one timed unit.	Bill 97110 ×2, 97112 ×2, G0283 ×1: $114.71, +29% versus a compliant version of the original.
P1 – Workers' Comp	The team is blocked while pulling data and formatting the PDF.	450 claims per day not sent (2,250 per week).	Sample 100 stuck claims, fix the largest reason, add tool checks and a checklist.
P3 – Claim not found	Claims the insurer never received or cannot match.	10,000 claims (about $1M, assumed).	Group by insurer, age and amount; sample; fix in groups; add a daily no-response check.

3. Problem 2 – Closed Encounters Analysis

Objective. Identify encounters (patient visits) closed in the client's EHR that were not imported into Normandy (Task 1), and explain why (Task 2). An encounter that is not imported is never billed.

3.1 Method

- **Cleaning (Python):** aligned column names, removed 2 blank rows, converted both date formats to dates, split the database code list into single codes, removed 331 repeated code lines, and aggregated both files to one row per encounter (patient, provider, date of service).
- **Analysis (SQL, PostgreSQL / Retool Database):** a left anti-join to find missing encounters, and grouped queries to compare missing *rates* rather than counts.
- **Reconciliation:** 6,612 EHR encounters = 6,330 imported + 282 missing (4.26%).

3.2 Task 1 – Missing encounters

The full list of 282 missing encounters (patient, date of service, provider, codes and reason) is provided in *p2_missing_encounters.csv* and in the dashboard.

3.3 Task 2 – Root cause

Eleven codes appear in the EHR but never in the database. None of them are valid CPT codes, which are five digits or G followed by four digits: NORCM (206 encounters), sp90 (36), SP (27), TOS115 (27), SP$90 (7), SP90 (7) and five single occurrences. They appear to be internal notes entered in the code field. *Assumption:* NORCM means the visit should not be billed through RCM, SP indicates self-pay, and TOS indicates payment at the time of service.

Segment	Encounters	Missing	Missing rate
Contains a non-CPT code	208	205	98.6%
Valid CPT codes only	6,404	77	1.2%
Total	6,612	282	4.3%

Rates matter here. Code 97140 appears in more missing encounters (258) than NORCM (203) simply because it is on 6,265 encounters; its missing rate is only 4.1%. All valid CPT codes have missing rates of roughly 2–5%, so no valid code is the cause.

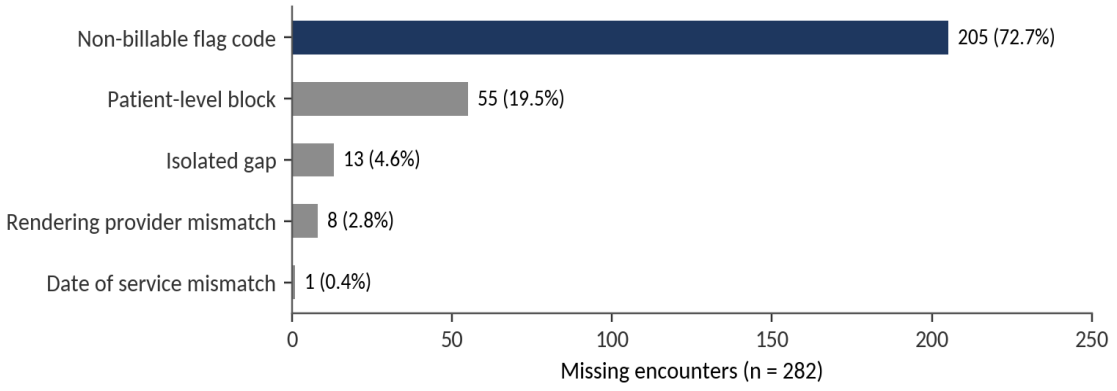


Figure 2. Missing encounters by reason.

- **Non-CPT codes (205, 72.7%)** are blocked by the import filter.
- **Patient-level blocks (55, 19.5%):** six patients are missing whole runs of visits (e.g. Layla Brown 20, Carter Lee 15). Monthly rates are flat (1.05%, 1.06%, 1.49%), which rules out a system outage; a patient or insurance setup issue is likely and should be confirmed.

- **Entry mismatches (9):** eight encounters exist in the database under a different provider and one with a date two days apart. **Isolated gaps:** 13.

Additional risks. 202 of the 208 blocked encounters also contain billable CPT codes, so any incorrectly entered flag results in lost revenue. Three flagged encounters were imported, which shows the filter is applied inconsistently. 108 encounters exist twice in the database, which creates a double-billing risk.

3.4 Recommendation

- Confirm the meaning of each non-CPT code with the client and maintain an approved exclusion list.
- Route excluded encounters that contain billable codes to a review queue instead of dropping them silently.
- Review the setup of the six blocked patients and correct the nine mismatched records.
- Monitoring: run the Task 1 anti-join daily and alert when a non-excluded encounter is missing for more than 48 hours. Target: less than 0.5%.

4. Problem 4 – Debugging Revenue

Medicare prices each CPT code from three components measured in Relative Value Units (RVUs): Work (clinician effort), Practice Expense (PE: rent, equipment, staff) and Malpractice (MP). Each is multiplied by a local cost index, the Geographic Practice Cost Index (GPCI), and the total is converted to dollars with the Conversion Factor (CF).

Payment = (Work RVU × Work GPCI + PE RVU × PE GPCI + MP RVU × MP GPCI) × CF

4.1 Part A – 97110 in localities 0210201 and 0730200

Locality 0210201 is Alaska (MAC 02102, locality 01) and 0730200 is Mississippi (MAC 07302, locality 00). Testing CMS data by year, the case amounts match calendar year 2023 exactly (2022 gives $39.26 and $27.69). In 2023, 97110 has Work 0.45, PE 0.42 and MP 0.01 RVUs, and the CF is $33.8872.

Locality	GPCI (W / PE / MP)	Work	PE	MP	Total RVU	Payment
Alaska	1.500 / 1.100 / 0.603	0.675	0.462	0.006	1.143	$38.73
Mississippi	1.000 / 0.847 / 0.720	0.450	0.356	0.007	0.813	$27.55

The code, RVUs and conversion factor are the same in both places; only the GPCIs differ. Alaska has a legislated work GPCI floor of 1.5 and higher practice costs. The work component alone accounts for $7.62 (about 68%) of the $11.18 difference.

4.2 Part B – 97140 paid ~$20 in claim #1 but ~$40 alone

Claim #1 contains four therapy codes on the same day, and all four carry CMS multiple-procedure indicator 5. Under the Multiple Procedure Payment Reduction (MPPR), the code with the highest PE is paid in full and all other codes receive 50% of their PE. Aetna applies the same rule to therapy services (policy effective October 2013).

Code (2023 RVUs)	Work	PE	MP	Treatment in claim #1
97112	0.50	0.50 (highest)	0.01	Paid in full
97110	0.45	0.42	0.01	PE reduced 50%
97140	0.43	0.37	0.01	PE reduced 50%
G0283	0.18	0.17	0.01	PE reduced 50%

Using national rates, 97140 alone pays (0.43 + 0.37 + 0.01) × 33.8872 = $27.45; in claim #1 it pays (0.43 + 0.185 + 0.01) × 33.8872 = $21.18, a 23% reduction. The difference is therefore a policy reduction, not an error. Aetna pays its own contracted rates (about $40 versus Medicare's $27), and its reduction (about 50%) is larger than Medicare's PE-only reduction. *Assumption:* Aetna's contract applies a deeper multiple-procedure reduction; this should be confirmed from the remittance reason code (CARC 59, processed under multiple-procedure rules). NCCI edits (code pairs requiring modifier 59/XS) do not apply to these four codes.

4.3 Part C – 8-minute rule

Under the CMS 8-minute rule (Claims Processing Manual, Chapter 5, §20.2), total *timed* minutes are converted to units (8–22 = 1, 23–37 = 2, 38–52 = 3, 53–67 = 4). Each code receives its full 15-minute units and any remaining unit goes to the code with the most leftover minutes. *Untimed* codes are billed once per visit. G0283 (unattended electrical stimulation) is untimed; Medicare does not accept 97014 for this service.

- Timed minutes: 97110 (38) + 97112 (24) = 62 minutes, which allows 4 units.
- Full units: 97110 = 2 units with 8 minutes left; 97112 = 1 unit with 9 minutes left.
- Remaining unit: 8 + 9 = 17 minutes gives one more unit, assigned to 97112 (9 minutes versus 8).

- G0283 is untimed, so 1 unit regardless of the 45 minutes.

Code	Billed	Correct	Issue
97110	2	2	None
97112	1	2	Under-billed by one unit
G0283	3	1	Over-billed by two units (compliance risk)

Version (2023 national rates, MPPR applied)	Allowed amount
As submitted (includes two non-allowable G0283 units worth $18.64)	$107.59
Compliant version of the original units	$88.95
Recommended: 97110 ×2, 97112 ×2, G0283 ×1	$114.71

The recommended claim is compliant and pays $25.76 (29%) more than a compliant version of the original. Medicare pays 80% of the allowed amount. To prevent recurrence, a pre-submission check should recalculate units from the documented minutes and flag untimed codes billed with more than one unit.

5. Problem 1 – Workers' Compensation Claims

Situation. Workers' Compensation (WC) claims cover patients injured at work and are paid by the employer's insurer. They must be mailed on paper. The offshore team uses an in-house tool to pull the claim data and format a PDF, which is then uploaded and mailed by a vendor. The team completes 50 of the expected 500 claims per day, leaving **450 claims per day (2,250 per week)** unsent. *Assumption:* at about $100 per claim, roughly $45,000 of payments are delayed each day.

The team is blocked while pulling data and formatting the PDF, before anything reaches the vendor. The vendor is therefore not a cause; its mailing time is tracked only after the fix.

5.1 Diagnosis

Two India operators record 100 stuck claims over one to three days, noting the step where each claim stopped (pull data, format PDF, upload) and the reason. The reasons are grouped, counted and ranked (Pareto) so that the largest cause is fixed first. Likely categories are missing WC information (claim number, date of injury, employer or insurer address), tool errors, missing attachments, unclear steps for operators, and upload rejections.

5.2 If the tool is the cause

- Validate required WC fields before generating the PDF and show a clear message (e.g. "Missing: date of injury").
- Pre-fill data the system already holds (patient, codes, provider) and log every error so failures can be counted automatically.
- Use the engineer's two hours per week only on the largest tool issue shown by the data.

5.3 If the workflow is the cause

- A one-page checklist of required WC fields, where to find them and what to do when one is missing.
- A 30-minute training session led by an experienced operator, and a 'stuck' queue with a reason field so no claim is left untracked.

5.4 Other possible causes

- The WC information was never captured by the clinic or never pulled into our system, similar to the missing encounters in Problem 2. This must be fixed upstream with the client.
- The upload step rejects the file, if completion includes the upload.

5.5 Metrics

Metric	Target
Claims completed per day	500
Share completed without getting stuck	Above 90%
Stuck claims by reason (weekly)	Decreasing
Minutes per claim	Decreasing
After the fix: vendor mailing time and days to payment	Within agreed time; decreasing

5.6 Resources and timeline

Period	India team (20)	Mountain View team (3)	Engineer (2 h/week)
Days 1–3	2 operators log 100 stuck claims; others continue	1 operator reviews the sample and calls the main WC insurers	Reviews tool error logs

Period	India team (20)	Mountain View team (3)	Engineer (2 h/week)
Week 1	Use the checklist and stuck queue	Delivers the training	Fixes the largest tool issue
Weeks 2–4	Return to full volume	Reviews the weekly figures	Fixes the next issue

Research tools: Google for WC required fields, one or two calls to major WC insurers, and ChatGPT to draft the checklist for review. *Assumption:* once fixed, a claim takes 10 minutes and an operator spends 7 hours per day on claims (42 claims each), so about 12 operators can complete 500 claims per day, releasing about 8 India operators for Problem 3. Only one of the three Mountain View operators is needed.

6. Problem 3 – Reconciliation: Claim Not Found

Situation. Claims were submitted, but when the team calls, the insurer cannot find them, so they remain unpaid. There are 10,000 such claims across 150 insurers (*assumption:* about $100 each, or roughly $1M). Insurers only accept claims up to a filing deadline, so older claims are the most urgent.

6.1 Prioritisation

- Combine all 10,000 claims in one table (claim ID, insurer, date submitted, amount).
- Group and count by insurer (a few insurers are likely to hold most of the claims), by age (oldest first) and by amount (largest first).

6.2 Diagnosis

Take about 20 claims from each of the five largest insurers and check whether each claim was actually sent, whether it went to the correct insurer, whether the patient's insurance ID is correct, and whether the insurer ever replied. Record and group the reasons; a finding such as "60 of 100 were sent to the wrong insurer" allows thousands of claims to be fixed at once.

6.3 Actions

Finding	Action
Claim was never sent	Submit it
Wrong insurer or ID	Correct and resubmit
Past the filing deadline, with proof of timely submission	Appeal with the proof
Small amount that cannot be recovered	Write off

6.4 Resources and capacity

Eight India operators (released from Problem 1) check claim status by phone or insurer website during US business hours. One Mountain View operator performs the sample review with me. The engineer is not needed initially; I build the tracker and daily report in SQL. *Assumption:* 6 claims per operator-hour and 6 hours per day gives 288 claims per day: about 35 working days if handled one by one, or about 21 working days (roughly four weeks) if 40% can be resolved in groups.

6.5 Prevention and metrics

- A daily list of claims submitted seven or more days ago with no response from the insurer, so they are followed up early.
- Validate insurer and patient ID before submission, and review a weekly count of 'not found' claims by insurer.
- Metrics: claims remaining (10,000 to 0), claims closed per day, dollars recovered, and claims lost to the filing deadline (target 0).

7. Assumptions

- Problem 2: meanings of NORCM, SP and TOS; a patient-level block is three or more missing visits for one patient; a date mismatch is within seven days.
- Problem 4: the case values correspond to CMS calendar year 2023; Aetna applies a deeper multiple-procedure reduction than Medicare; national rates are used in Parts B and C.
- Problems 1 and 3: about $100 per claim; 10 minutes per WC claim; 6 status checks per operator-hour; 40% of Problem 3 resolvable in groups. These would be replaced with real data in the first week.

8. References

[1] Centers for Medicare & Medicaid Services (CMS). Physician Fee Schedule datasets – RVUs and GPCIs, CY2022–2023. pfs.data.cms.gov

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[3] CMS. Locality Key. cms.gov/files/document/cy-2022-locality-key.pdf

[4] CMS. Medicare Claims Processing Manual, Chapter 5, §20.2; APTA, Coding for Timed Codes. apta.org

[5] American Physical Therapy Association (APTA). Multiple Procedure Payment Reduction. apta.org

[6] Aetna Multiple Procedure Reduction Policy for therapy services (via Midlands Choice). midlandschoice.com

[7] WebPT. Modifier 59 and NCCI edits for physical therapy. webpt.com

[8] SPRY. G0283 vs 97014 billing for Medicare. sprypt.com

[9] Retool. Retool Database documentation. docs.retool.com

Supporting files: P2_Step1_Cleaning.ipynb (Python cleaning), p2_task1_task2.sql (SQL analysis), p2_missing_encounters.csv (Task 1 list).